

Linux file System Hierarchy and Scenario-Based Practice

Part 1 - Linux file system Hierarchy

/ (Root directory) →

- starting point of the entire file system.
- Everything exist under /

ls -l / → shows home, etc, var, bin, etc.

When to use? → I would use this when I want to navigate the entire system.

/home →

- stores user directories.

ls -l /home →

When to use? → To access user files.

/root →

Home directory of root user (admin)

sudo ls -l /root

When to use? when performing admin level task.

4. /etc →

- stores configuration files.

ls -l /etc

When to use? To modify system or service configuration.

5. /var/log →

- stores log files.

ls -l /var/log

When to use? → to troubleshoot errors and check logs.

6. /tmp →

- Temporary files

ls -l /tmp

When to use? → for temporary storage during testing

7. /bin →

- Essential system commands.

ls -l /bin

When to use → to locate core system commands.

8. /usr/bin →
• user installed binaries

When to use? → to find installed software commands.

9. /opt →
• Third party applications.

When to use? → used for manually installed appⁿ.

* Hands - On tasks -

1. find largest log files -

```
du -sh /var/log/* 2>/dev/null | sort -h | tail -5
```

du -sh → size of files

sort -h → sorts by size

tail -5 → shows largest 5

2. check config file -

cat /etc/hostname → shows system hostname

2. check home directory -

ls -la ~ →

list your files including hidden ones.

Part 2 - Scenario - Based Practice

- Scenario 1 : Service Not starting
 step 1 - systemctl status myapp
 ↳ checks if service is running / failed.

step 2 -
 journalctl -u myapp -n 50
 ↳ check error logs

step 3 -
 systemctl is-enabled myapp
 ↳ check if it starts on boot.

step 4 -
 systemctl restart myapp
 ↳ Try restarting after checking logs.

- Scenario 2 : High CPU Usage
 step 1 -
 top → see live CPU usage

step 2 →
 ps aux --sort=-%CPU | head -10
 ↳ find top CPU-consuming processes

step 3 →
 ps -p <PID> -o pid,cmd,%cpu,%mem
 ↳ inspect specific process.

- Scenario 3: finding logs

step 1 -

`systemctl status ssh`
↳ confirms service exists

step 2 -

`journalctl -u ssh -n 50`
↳ view recent logs

step 3 -

`journalctl -u ssh -f`
↳ follow logs live

Scenario 4: Permission Denied

step 1:

`ls -l /home/user/backup.sh`
↳ check permissions

step 2 -

`chmod +x /home/user/backup.sh`
↳ add execute permission

step 3 -

`ls -l /home/user/backup.sh`
↳ verify permission change

step 4 -

`./backup.sh`
↳ Run script